Muhammad Nur Shahril Iskandar

m.n.shahril.iskandar@gmail.com | linkedin.com/in/shahril-iskandar | github.com/Shahril-Iskandar

Education

Griffith University — Australia
Australian Centre for Precision Health and Technology (PRECISE), School of Allied Health, Sport and Social Work — 2025-Present

- Doctor of Philosophy (PhD) in Biomechanics
- Investigating personalised gait modifications to alter hip joint loading in people with hip osteoarthritis.
- Integrating 3D motion analysis, electromyography, and MRI-informed neuromusculoskeletal modelling to estimate hip joint contact forces.

Nanyang Technological University (NTU) — Singapore
Department: Physical Education and Sports Science Academic Group — Jun 2023

- B.S., Sport Science and Management with Honours (Highest Distinction)

Research Experience

Senior Research Assistant, Nanyang Technological University — Singapore
Sports Biomechanics Lab | Principal Investigators: Phillis Teng, PhD & Kong Pui Wah, PhD — Aug 2023 - Feb 2025

- Contributed to research investigating biomechanical risk factors for early knee osteoarthritis and adapted table tennis coaching for individuals with physical disabilities.
- Conducted a systematic review on the diagnostic utility of ultrasonography for early knee osteoarthritis using Covidence [J8].
- Investigated markerless motion capture for biomechanical analysis in non-standardised sporting environments [C3, CA4].
- Assisted with research design, ethics applications, and mentoring undergraduate students during data collection.

Research Intern, Agency for Science, Technology and Research, Bioinformatics Institute — Singapore
Biophysical Modelling Lab | Principal Investigator: Chiam Keng Hwee, PhD — Jan 2023 - Jun 2023

- Contributed to research on data-driven gait rehabilitation for individuals with lower-limb amputations.
- Implemented generative AI using Stable Diffusion to enhance existing open-source pose estimation algorithm (OpenPose & DeepLabCut) in identifying lower limb amputee's anatomical landmarks for gait analysis [J5].
- Developed Python scripts for biomechanical data processing and analysis.

Undergraduate Research Assistant, Nanyang Technological University — Singapore
Sports Biomechanics Lab | Principal Investigator: Kong Pui Wah, PhD — Apr 2022 - May 2023

- Investigated the biomechanical effects of passive exoskeletons during military load carriage [CA2, CA3, J2, J4].
- Developed MATLAB and Python scripts to extract data and conduct data analysis of ground reaction forces using statistical parametric mapping, spm1d.

Undergraduate Research Programme (URECA), Nanyang Technological University — Singapore
Sports Biomechanics Lab | Principal Investigator: Kong Pui Wah, PhD — Aug 2021 - Aug 2022

- Developed a video-based approach to assess running biomechanics, including rearfoot kinematics and gait patterns [C1, C2, J1, J3].
- Facilitated over 40 participants' recruitment and utilized Kinovea to analyze running kinematics.

Undergraduate Research Assistant, Nanyang Technological University — Singapore
Human Bioenergetics Lab | Principal Investigator: Yang Yifan, PhD — Sep 2020 - Mar 2021

- Supported research investigating the effects of leucine supplementation on muscle maintenance during weight loss, including participant recruitment, anthropometric assessments, and data management.

Publications

* indicates corresponding author, † indicates equal contribution

Peer-reviewed Journal Articles

[J8] **Iskandar, M. N. S.**, Teng, P.*, Mukherjee, S. & Tan, S. Y. (Accepted). Utility of ultrasound in the diagnosis of early knee osteoarthritis: A systematic review. *Sports Medicine and Arthroscopy Review*.

[J7] Loh, R. B. C, Pan, J. W., **Iskandar, M. N. S.**, & Kong, P. W.*(2026). Predicting running-related injuries from functional, kinetic and kinematic data. *International Journal of Sport Medicine*. 47(05):372-380. doi: 10.1055/a-2706-5516

[J6] Marican, M. A., Chandra, L., Tang, Y., **Iskandar, M. N. S.**, Lim, C. X. E. & Kong, P. W.*(2025). Biomechanical Effects of a Passive Back-Support Exosuit during Simulated Military Lifting Tasks - An EMG Study. *Sensors*. 25(10), 3211. doi: 10.3390/s25103211

[J5] Zhou, T†, **Iskandar, M. N. S.**†, & Chiam, K. H.* (2025). Diffusion Models Enable Zero-Shot Pose Estimation for Lower-Limb Prosthetic Users. *PLOS Digital Health*. 4(3), e0000745. doi: 10.1371/journal.pdig.0000745

[J4] Kong, P. W.*, Koh, A. H., Ho, M. Y. M., **Iskandar, M. N. S.**, & Lim, C. X. E. (2024). Effectiveness of A Passive Military Exoskeleton in Offloading Weight during Static and Dynamic Load Carriage: A Randomised Cross-Over Study. *Applied Ergonomics*. 119, 104293. doi: 10.1016/j.apergo.2024.104293

[J3] **Iskandar, M. N. S.**, Loh, R. B. C., Ho, M. Y. M., Pan, J. W. & Kong, P. W.* (2023). Crossover Gait in Running and Measuring Foot Inversion Angle at Initial Foot Strike: A Front-View Video Analysis Approach. *Frontiers in Bioengineering and Biotechnology*. 11, 1210049. doi: 10.3389/fbioe.2023.1210049

[J2] Kong, P. W.*, **Iskandar, M. N. S.**, Koh, A. H., Ho, M. Y. M., & Lim, C. X. E. (2023). Validation of In-Shoe Force Sensors During Loaded Walking In Military Personnel. *Sensors*. 23(14), 6465. doi: 10.3390/s23146465

[J1] Pan, J. W., Ho, M. Y. M., Loh, R. B. C., **Iskandar, M. N. S.**, & Kong, P. W.* (2023). Foot Morphology and Running Gait Pattern between the Left and Right Limbs in Recreational Runners. *Physical Activity and Health*, 7(1), 43–52. doi: 10.5334/paah.226

Peer-reviewed Conference Proceedings

[C3] **Iskandar, M. N. S.**, & Teng, P. S. P.* (2024). Kinematics Comparison of OpenCap and IMU with Marker-Based Motion Capture In Treadmill Running: A Pilot Study. *Proceedings of the 42nd International Society of Biomechanics in Sports Conference.*

[C2] Loh, R. B. C., Ho, M. Y. M., **Iskandar, M. N. S.**, & Kong, P. W.* (2024). Two-Dimensional Kinematics Differences Between Sexes In Runners With and Without Patellofemoral Pain. *Proceedings of the 42nd International Society of Biomechanics in Sports Conference.*

[C1] **Iskandar, M. N. S.**, Loh, R. B. C., Ho, M. Y. M., Pan, J. W., & Kong, P. W.* (2022). Comparison of Rearfoot Inversion Angle at Initial Footstrike Measured From Front And Back View Videos. *Proceedings of the 40th International Society of Biomechanics in Sports Conference*, 40(1), 291.

Conference Abstracts

[CA5] Teng, P. S. P., Mukherjee, S., Tan, S. Y., **Iskandar, M. N. S.**, Lim, M. Y. K., See, R. Y. & Lim, B. W. B. (2025). Plantar Pressure Differences Between Low and High-Risk Knee Osteoarthritis Long-Distance Recreational Runners. *30th International Society of Biomechanics Conference. July 2025.*

[CA4] **Iskandar, M. N. S.**, Quek, B. T. L., Liu, H., Ma, C. M. S., & Kong, P. W. Feasibility of Markerless Motion Capture in Non-Standardised Conditions - A Case Study on a Table Tennis Player with Physical Disabilities. *10th Asian Society of Sport Biomechanics Conference*. December 2024.

[CA3] Lim, C., Kong, P. W., Koh, A. H., Ho, M., & **Iskandar, M. N. S.**. The physiological and biomechanical effects of a full-body passive exoskeleton on military load carriage. *6th International Congress on Soldiers' Physical Performance*. September 2023.

[CA2] **Iskandar, M. N. S.**, Koh, A. H., Ho, M. Y. M., Lim, C. X. E., & Kong, P. W. Validation of the loadsol® in-shoe force sensors during walking in military boots under heavy load carriage. *9th Asian Society of Sport Biomechanics Conference*. August 2023.

[CA1] Loh, R. B. C., Ho, M. Y., **Iskandar, M. N. S.**, Pan, J. W., & Kong, P. W. Reliability of video-based running gait analysis in recreational runners. *XXII International Conference on Mechanics in Medicine and Biology*. August 2022.

Conference Oral Presentations

[CP4] Feasibility of Markerless Motion Capture in Non-Standardised Conditions - A Case Study on a Table Tennis Player with Physical Disabilities, *10th Asian Society of Sports Biomechanics (ASSB) Conference*, Kuala Lumpur, Malaysia, December 2024

[CP3] Validation of the loadsol® in-shoe force sensors during walking in military boots under heavy load carriage, *9th Asian Society of Sports Biomechanics (ASSB) Conference*, Bangkok, Thailand, August 2023

[CP2] Two-Dimensional Video Analysis of the Rearfoot Inversion Angle at Initial Footstrike in Treadmill Running, *10th International Conference of Undergraduate Research (ICUR)*, Virtual presentation, September 2022

[CP1] Comparison of Rearfoot Inversion Angle at Initial Footstrike Measured From Front And Back View Videos, *40th International Society of Biomechanics in Sports (ISBS) Conference*, Liverpool, United Kingdom, July 2022

Teaching Experience

Tutor - Introductory Biomechanics (2011AHS) 2026

Community Service

Student Mentor 2025

Queensland Academies for Health Sciences and Griffith University

Mentored Year 10 students in developing innovative solutions to healthcare challenges through the nine-week Change in Action program, providing research guidance and feedback.

Awards

Anugerah Cemerlang MENDAKI Award 2023

Recognised for graduating with First Class Honours (Highest Distinction).

Internship Commendation Award 2023

Awarded to the top 15% of NTU internship students.

Best Thesis Oral Presentation Award 2022

Awarded at the 11th NTU Lau Teng Chuan Physical Education & Sports Science Symposium.

ISBS Student Travel Grant 2022

Travel support to attend the 40th ISBS Conference.

Skills

Programming: Python, MATLAB, R, Excel VBA, HTML/CSS

Software: SPSS, JASP, OpenSim, Visual3D, VICON Nexus

Languages: English (Native), Malay